Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2026

Course: B.Tech
Code: 155609

Subject: Introduction to Industry 4.0

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Which of the following is considered a key characteristic of Industry 4.0?
(i) Isolated business processes
(ii) Manual production systems
(iii) Smart and connected manufacturing
(iv) Paper-based communication

(b) The main objective of LEAN Production Systems is to
(i) Increase inventory levels
(ii) Increase machine downtime
(iii) Reduce waste and improve efficiency
(iv) Maximize paperwork

(c) A Cyber Physical System combines
(i) Hardware only
(ii) Physical processes with computational systems
(iii) Human resources and finance
(iv) Marketing and sales

(d) Which technology is primarily used to create secure and tamper-resistant transaction records?
(i) Blockchain
(ii) Big Data
(iii) Artificial Intelligence
(iv) Cloud Computing

(e) Cloud computing enables organizations to
(i) Eliminate all hardware requirements
(ii) Work without data storage
(iii) Access computing resources over the internet
(iv) Avoid cybersecurity measures

(f) Microsoft Azure is widely used for
(i) Automobile production
(ii) Industrial cloud solutions and data analytics
(iii) Circuit fabrication only
(iv) Manufacturing physical sensors

(g) An industrial application of AR is
(i) Payroll management
(ii) Data encryption
(iii) Virtual maintenance guidance
(iv) Inventory billing

(h) The Industrial Internet of Things mainly connects
(i) Industrial machines and devices
(ii) Mobile games
(iii) Television networks
(iv) Social media accounts

(i) A major cybersecurity challenge in Industry 4.0 is
(i) Improved automation
(ii) Reduced data generation
(iii) Increased connectivity leading to cyber threats
(iv) Lower productivity

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(j) UAV stands for

(i) Unified Automated Venture
(ii) Unmanned Aerial Vehicle
(iii) Universal Access Vehicle
(iv) Unmanned Automated Verification

Q.2 (a) Define Industry 4.0 and explain the major characteristics of the Fourth Industrial Revolution. [7]

(b) Explain how globalization has influenced the adoption of Industry 4.0 technologies across various industrial sectors. Illustrate your answer with suitable examples. [7]

Q.3 (a) Explain how Artificial Intelligence, Big Data and Blockchain technologies contribute to the realization of Industry 4.0. Support your answer with examples. [7]

(b) Describe the architecture and components of a Cyber Physical System in details. [7]

Q.4 (a) Define cloud computing and explain the concepts of IaaS, PaaS and SaaS in the context of Industry 4.0 applications. [7]

(b) Analyze the advantages and challenges of cloud adoption in Industry 4.0 environments with respect to scalability, security and cost optimization. [7]

Q.5 (a) Discuss how AR technologies assist industrial workers in maintenance, assembly, training and quality inspection processes. [7]

(b) Define Augmented Reality and Virtual Reality. Explain the major hardware and software components used in AR systems. [7]

Q.6 (a) Describe the architecture of Industrial Internet of Things systems and explain the key enablers of IIoT. [7]

(b) Explain how IIoT technologies facilitate industrial automation, process monitoring and data-driven decision-making. [7]

Q.7 (a) Discuss real-world applications of Industry 4.0 technologies in the oil, chemical, manufacturing and pharmaceutical industries. [7]

(b) Explain the role of Unmanned Aerial Vehicles in industrial inspection, surveillance, logistics and disaster management applications. [7]

Q.8 (a) A manufacturing industry experiences frequent machine failures. Demonstrate how AI and Big Data Analytics can be applied to implement predictive maintenance. [7]

(b) Evaluate the challenges of AR implementation in industrial environments and design an AR-based maintenance support system for a smart factory. [7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) Analyze the differences between traditional manufacturing systems and smart connected enterprises

(b) Compare 8 points of AWS, Azure and GCP for a smart manufacturing system

(c) Analyze the major cybersecurity threats and vulnerabilities associated with IIoT infrastructures in Industry 4.0

(d) Analyze and compare Industry 4.0 implementation strategies adopted by manufacturing and pharmaceutical industries. Discuss key benefits and challenges

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